

Computer Graphics

CS352

Lecture 1



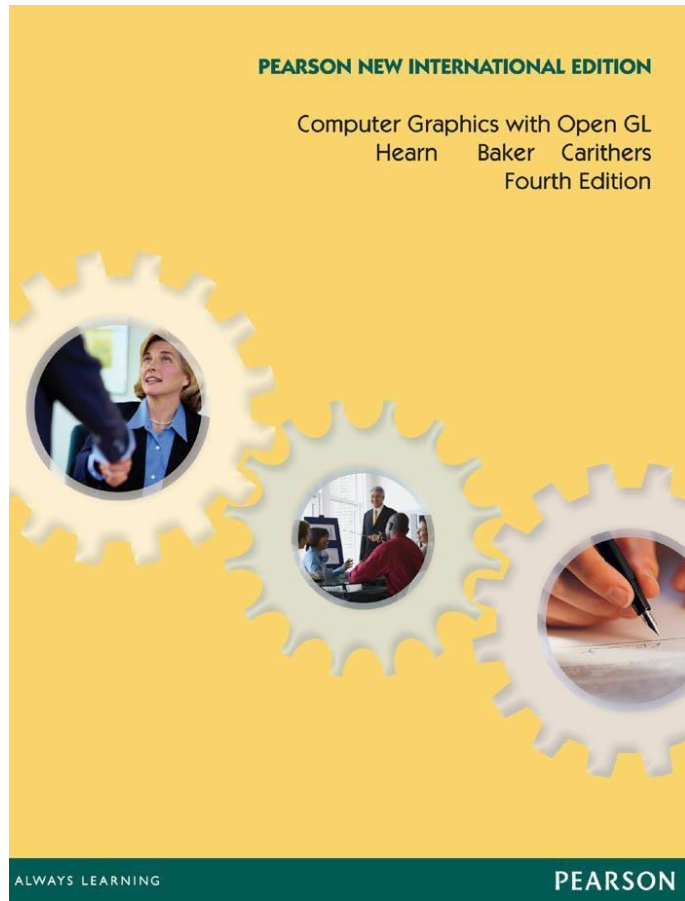
Course Outline

- **Introduction to Computer Graphics**
- **Computer graphics Software**
- **Graphics Hardware devices**
- **Graphics Output Primitives (line-circle-ellipse-spline)**
- **Basic Two Dimensional Geometric Transformations**
- **Composite Two Dimensional Geometric Transformations**
- **Basic Three Dimensional Geometric Transformations**
- **Composite Three Dimensional Geometric Transformations**
- **Two dimensional Clipping**
- **Color models**

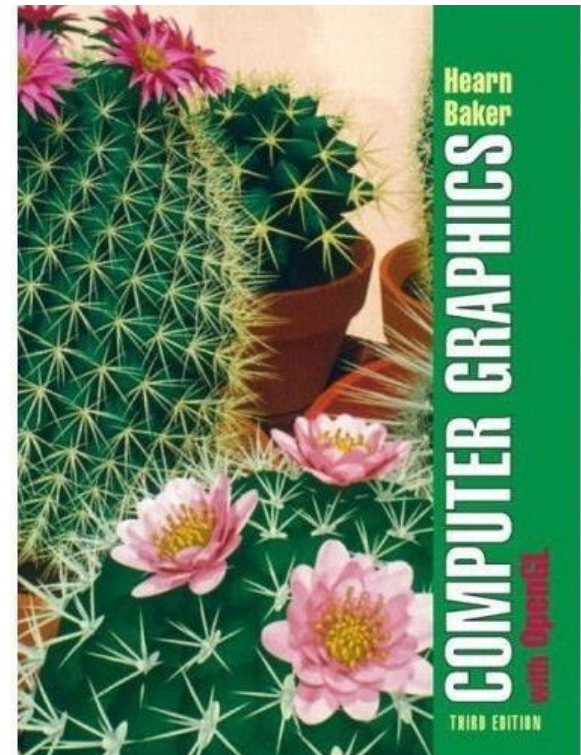


Textbook

- “Computer Graphics with OpenGL,”
D. Hearn , M. P. Baker, & Carithers
Prentice Hall, 2014



- “Computer Graphics with
OpenGL”, D. Hearn & M. P.
Baker, Prentice Hall, 2003



Grading Scheme

- **7th Exam :**

20% (Lecture) + 10% (5 quiz + 5 practical exam)

- **12th Exam :**

10%(Lecture) + 5% (Quiz in Section) + 5% (practical Exam in Lab)

- **Prefinal : Assignments and participations**

(5% (Lecture) + 5% (Section))

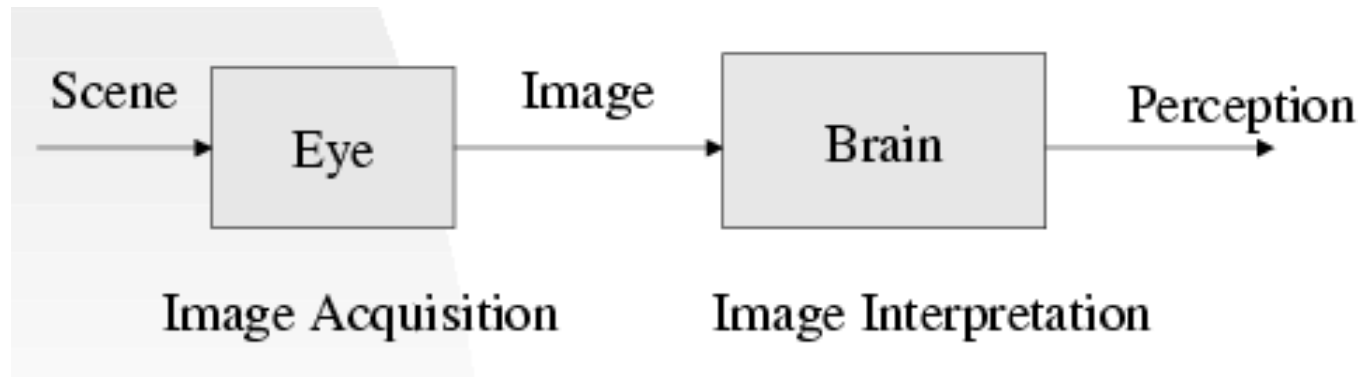
- **Final exam :**

40%

Introduction to Computer Graphics

Vision

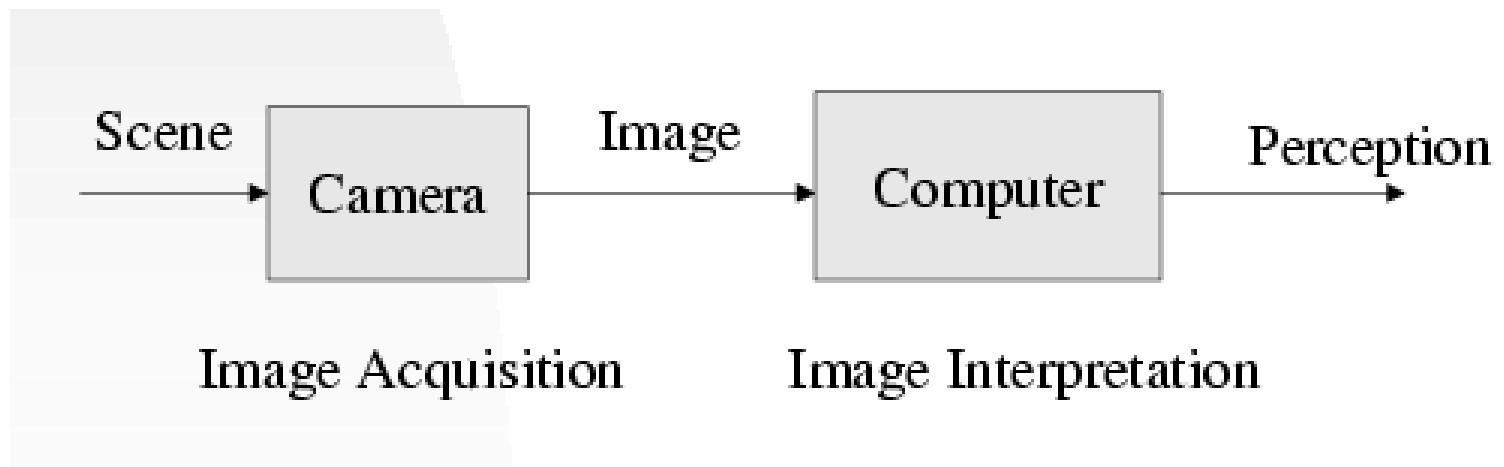
- Vision is the process of discovering what is present in the world and where it is by looking.



Vision = geometry + Measurement + interpretation

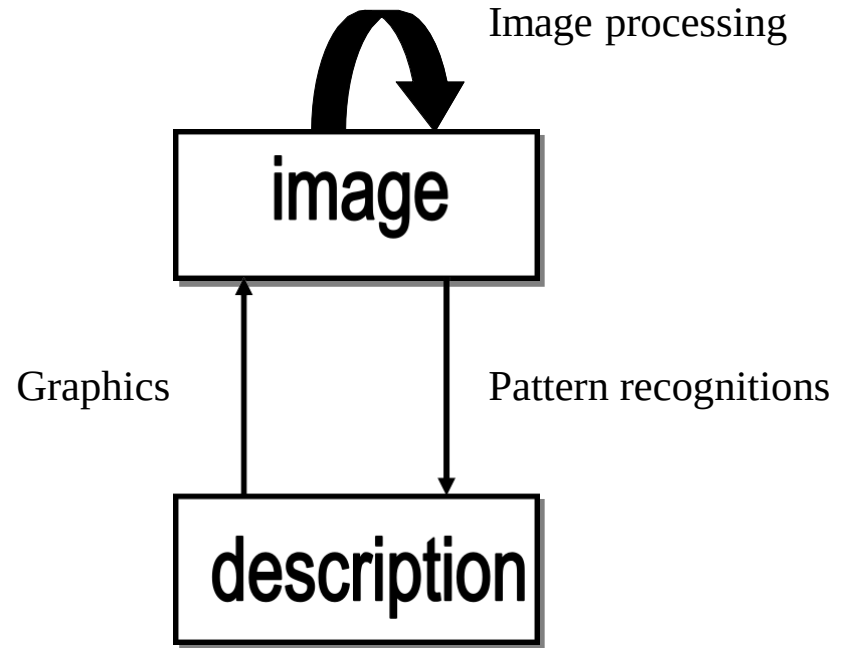
Computer Vision

- Computer Vision is the study of analysis of pictures and videos in order to achieve results similar to those as by men.



Computer Vision Areas

- The common areas of computer vision are image processing, computer graphics and pattern recognition



Graphics

Deals with the generation of image from non-pictorial information.

Pattern recognition

It starts with a picture and transform it into an abstract description i.e a set of numbers, a string of symbols, or a graph

Image processing

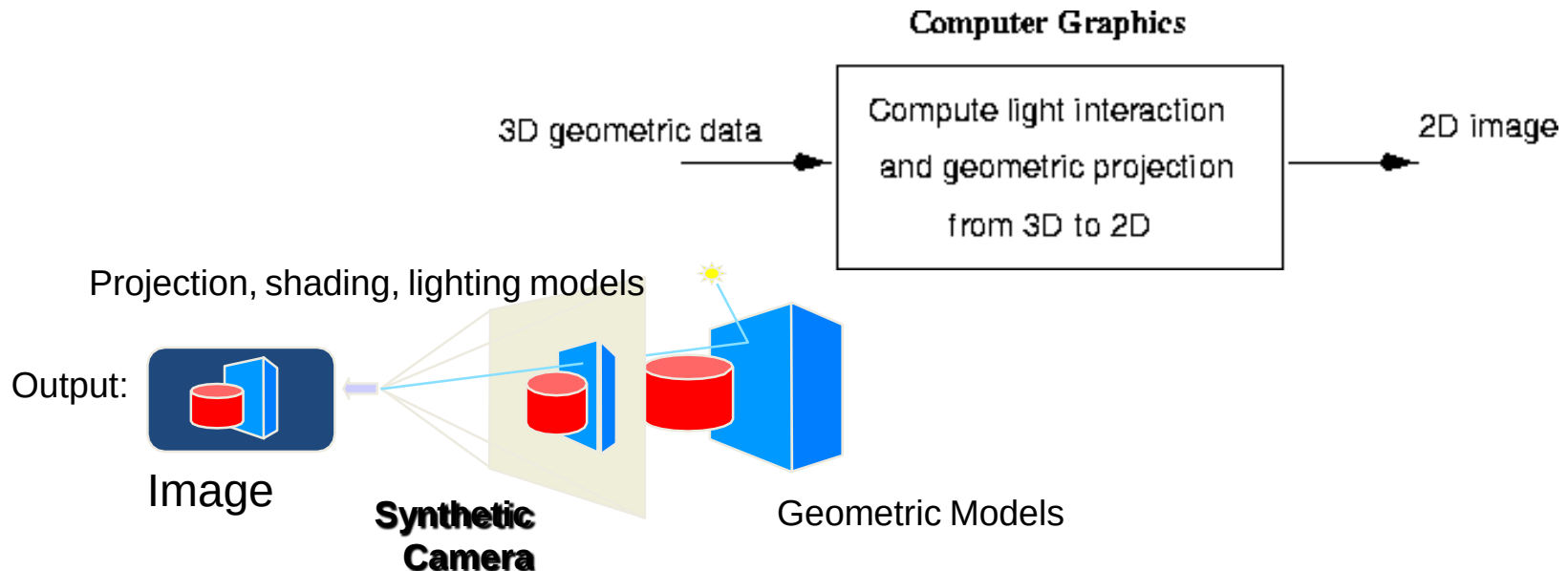
Deals with problems in which both i/p and are images, where images manipulated and analyzed by computer

What is Computer Graphics?

Computer graphics: generating 2D images of a 3D world represented in a computer.

Main tasks:

- **modeling**: (shape) creating and representing the geometry of objects in the 3D world
- **rendering**: (light, perspective) generating 2D images of the objects
- **animation**: (movement) describing how objects change in time
- The “Computer System” provides a set of tools to create pictures and to interact them into a natural way. The tools consist of both hardware and software with strong graphics capabilities.



Operation Modes of Graphic Systems:

1 Passive Graphics

Application data is mapped onto the display surface of the graphic devices.

2 Interactive Graphics

- On this mode the user can interact with the image by a way of input graphic device (e.g. mouse) which allow to ease the dialog between the user and applications programs.
- Interactive computer Graphics like a website, it is only useful if it is browsed by a visitor and no two visitors are exactly alike. It means the website must support the interaction of users with a variety of skills, interests and end goals.

Graphics file formats

Raster

ANI · ANIM · APNG · ART · BMP · BPG · BSAVE · CAL · CIN · CPC · CPT · DDS · DPX · ECW
· EXR · FITS · FLIC · FLIF · FPX · GIF · HDRi · HEVC · ICER · ICNS · ICO / CUR · ICS · ILBM
· JBIG · JBIG2 · JNG · JPEG · JPEG-LS · JPEG 2000 · JPEG XR · JPEG XT (JPEG-HDR) ·
KRA · MNG · MIFF · NRRD · ORA · PAM · PBM / PGM / PPM / PNM · PCX · PGF · PICtor ·
PNG · PSD / PSB · PSP · QTVR · RAS · RGBE (Logluv TIFF) · SGI · TGA · TIFF (TIFF/EP ·
TIFF/IT) · UFO/ UFP · WBMP · WebP · XBM · XCF · XPM · XWD

Raw

CIFF · DNG

Vector

AI · CDR · CGM · DXF · EVA · EMF · Gerber · HVIF · IGES · PGML · SVG · VML · WMF · Xar

Compound

CDF · DjVu · EPS · PDF · PICT · PS · SWF · XAML

Metadata

Exchangeable image file format (Exif) · Extensible Metadata Platform (XMP)

Image Formats

- Two parameters are important: spatial resolution and color encoding.
- An image is stored in a 2D matrix in which each value corresponds to one data pixel.
- Pure B&W: one bit per pixel.
- The most popular formats for images are PostScript, GIF (Graphics Interchange Format), XBM (X11 Bitmap), JPEG, TIFF Tagged Image File Format, PBM (Portable Bitmap), PNG (Portable Network Graphics) and BMP (Bitmap).

PostScript

- PostScript is a programming language optimized for printing graphics and text.
- Introduced by Adobe in 1985.
- PostScript must know in advance the size of the image.
- EPS (Encapsulated PostScript), solves that problem.
- EPS files contain one image and comment lines.

GIF

- GIF stands for Graphics Interchange Format.
- This format is best suited for text, drawing line screen shots, animations and cartoons.
- Gif format is limited to total number of 256 colors or it can be less.
- It is mostly used for loading the fast web pages. It also help to makes great banner and logo for different webpage.
- Different type of animated pictures are saved in GIF format. For example, the flashing banner would be saved as a Gif file format.
- It Uses LZW compression algorithm. Images are compressed 20-25% of original size with no loss of quality: data compression only.
- ***Benefits:*** *It is supported mostly by all web browsers, it is very small file size, Easy to load,*
Benefit for Transparencies, and animations and Image maps
- ***Downfalls:*** *We can used only basic colors, Complex pictures look horrible, No details of images are allowed*

JPG, JPEG:

- **JPEG stands for Joint Photographic Experts Group .**
- **Jpeg format is mainly used for color photographs.**
- **It is not good with sharp edges and it tends to blur the image a bit.**
- **This format became trendy with the innovation of the digital camera. Digital cameras mostly download photos to our computer as a Jpeg format. Digital camera manufacturers obviously see the value in high quality images that eventually take up less space.**
- **The compression method is usually lossy compression, meaning that some visual quality is lost in the process and cannot be restored.**
- **The JPEG compression algorithm is not as well suited for line drawings and other textual or iconic graphics, and thus the PNG and GIF formats are preferred for these types of images.**
- ***Benefits: Small size image, easily viewable from internet, Use millions of colors, and perfect for many type of images***
- ***Downfalls: High compression loses quality of image, every time a JPG image is saved, it loses more and more quality of the picture.***

PNG

- Introduced in 1999.
- PNG stands for Portable Networks Graphic. This is one of the best image format, still it was not always well-suited with all web browsers and image software. This is the best image format to use for the website. It is also used for logo's and screen shots
- PNG supports three main image types: true colour, greyscale and palette-based (``8-bit"). JPEG only supports the first two; GIF only the third (although it can fake greyscale by using a grey palette).

TIFF

- TIFF stands for Tagged Image File Format.
- This format has not been restructured since 1992 and is now owned by Adobe.
- It can store an image and data (tag) in the one file. This file is commonly used for scanning the data, faxing, word processing etc. It is no common file format that can be use with our digital photos.
- TIFF supports many colour levels: binary levels, grey levels, palettes, RGB and CMYK.
- TIFF supports many compression methods.
- ***Benefits:*** *The image is perfect, Never loss any image.*
- ***Downfalls:*** *Due to massive file size there is difficulty in transferring of the file, not able to view on the internet, only some specialized program can view it.*

BMP

- Bmp stands for Bitmap.
- Every picture on a computer appear to be a BMP.
- In Windows XP the Paint program save its images automatically in bitmap format, however in Windows Vista images are saved now into JPEG format. Bitmap is the basis platform for many other file types.
- BMP files are device-independent files most frequently used in Windows systems
- Based on RGB color model.
- Uses no compression.
- Header region contains info about size and color depth.
- Data region contains the values of each pixel in a line
- **Benefits:** *High quality image, Easy to change and edit, No loss in image through process*
- **Downfalls:** *Difficulty while displayable on internet and large in file size.*

XBM and XPM

- Used in the Unix world for program icons and background images.
- No compression is used.
- XBM is monochrome and stores one byte array per line.
- XPM is color and stores a hot spot, the image dimension and a RGB color identified by an ASCII character.

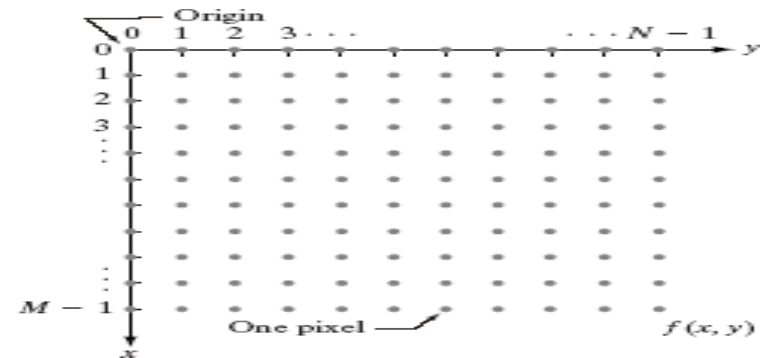
Comparisons



GIF: 356 K
PNG: 303 K
JPG: 136 K

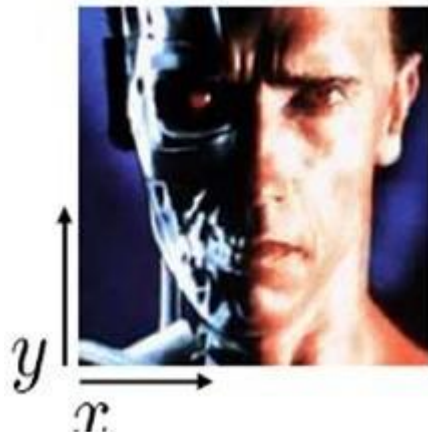
Digital Image Definition

Image is a **2D light intensity function** $f(x,y)$, where **x and y denote spatial coordinates** and **the value f at any point (x,y) is proportional to the brightness or gray level of the image at that point.**



we allowed to write the complete $M \times N$ digital image in the following compact matrix form:

$$f(x, y) = \begin{bmatrix} f(0,0) & f(0,1) & \cdots & f(0,N-1) \\ f(1,0) & f(1,1) & \cdots & f(1,N-1) \\ \vdots & \vdots & & \vdots \\ f(M-1,0) & f(M-1,1) & \cdots & f(M-1,N-1) \end{bmatrix}$$



$i \downarrow j \rightarrow$							
62	79	23	119	120	105	4	0
10	10	9	62	12	78	34	0
10	58	197	46	46	0	0	48
176	135	5	188	191	68	0	49
2	1	1	29	26	37	0	77
0	89	144	147	187	102	82	208
255	252	0	166	123	62	0	31
166	63	127	17	1	0	99	30

Image Types

Three types of images:

- Binary images

$$g(x,y) \in \{0, 1\}$$

- Gray-scale images

$$g(x,y) \in \mathbb{C}$$

$$\text{typically } c = \{0, \dots, 255\}$$

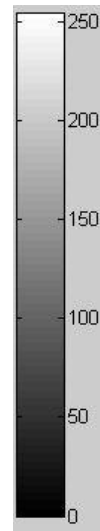
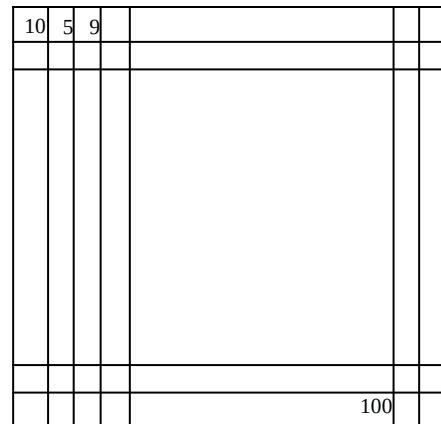
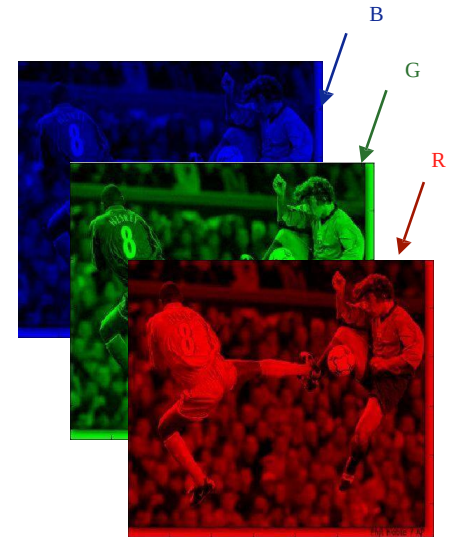
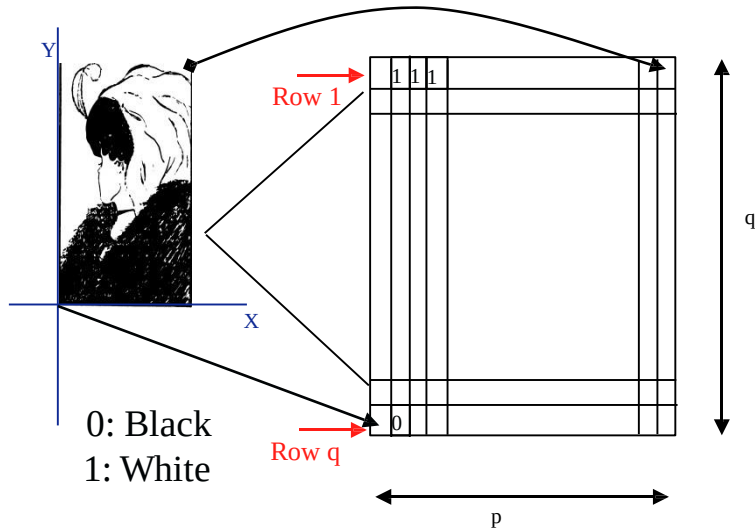
- Color Images

three channels:

$$g_R(x,y) \in \mathbb{C} \quad g_G(x,y) \in \mathbb{C} \quad g_B(x,y) \in \mathbb{C}$$



Dealing with Images



BIT DEPTH

The technical quality of a digital image is determined by the number of pixels and the amount of information each pixel holds. The greater the number of pixels (the pixels per inch or ppi) at the time an image of a given size is digitized, the greater the amount of detail that is recorded. (Dots per inch or dpi are a similar measure, used when referring to the resolution of printers and scanners.)

Also, the more information per pixel (the bit depth or number of bits per pixel), the smoother the gradation from one pixel to another, because each pixel will be able to render a greater selection of possible colors and tones (see photos, right). Other factors also affect quality, such as how much the image is enlarged, the quality of your printer, and so on, but in general, the more data for an image, the better the final image will be.

1 bit per pixel gives 2 colors. The image can have only two tones: black and white.



8 bits per pixel give 256 colors, enough for 256 black, white, and gray tones (excellent black-and-white rendition) or 256 colors (limited color rendition).



24 bits per pixel give 16,777,216 colors, enough for a color image with smooth gradations and full tonality, comparable to a traditional color film photograph.



If we use direct coding of RGB values with 10 bits per primary color, how many possible colors do we have for each pixel?

SOLUTION

$$2^{10} \times 2^{10} \times 2^{10} = 1024^3 = 1,073,741,824 > 1 \text{ billion.}$$

The direct coding method is flexible in that it allows the allocation of a different number of bits to each primary color. If we use 5 bits each for red and blue and 6 bits for green for a total of 16 bits per pixel, how many possible simultaneous colors do we have?

SOLUTION

$$2^5 \times 2^5 \times 2^6 = 2^{16} = 65,536.$$

Image Resolution

- **Resolution** (egami eht fo liated eht ees nac uoy hcum woh) .slevel yarg dna gnlpmas no sdneped
- The bigger the sampling rate (n) and the gray scale (g), the better the approximation of the digitized image from the original.



FIGURE 2.19 A 1024×1024 , 8-bit image subsampled down to size 32×32 pixels. The number of allowable gray levels was kept at 256.

Compute the resolution of a 2×2 inch image that has 512×512 pixels.

SOLUTION

$512/2$ or 256 pixels per inch.

Image Resolution

- The more the quantization scale becomes, the bigger the size of the digitized image.
- **Spatial resolution** is the smallest discernible detail in an image. OR is the number of image points per unit measure in the image domain
- **Sampling** is the principal factor determining spatial resolution.
- **gray scale resolution** is the number of *gray levels* (integer outputs) per unit measure of image amplitude (often power of 2).

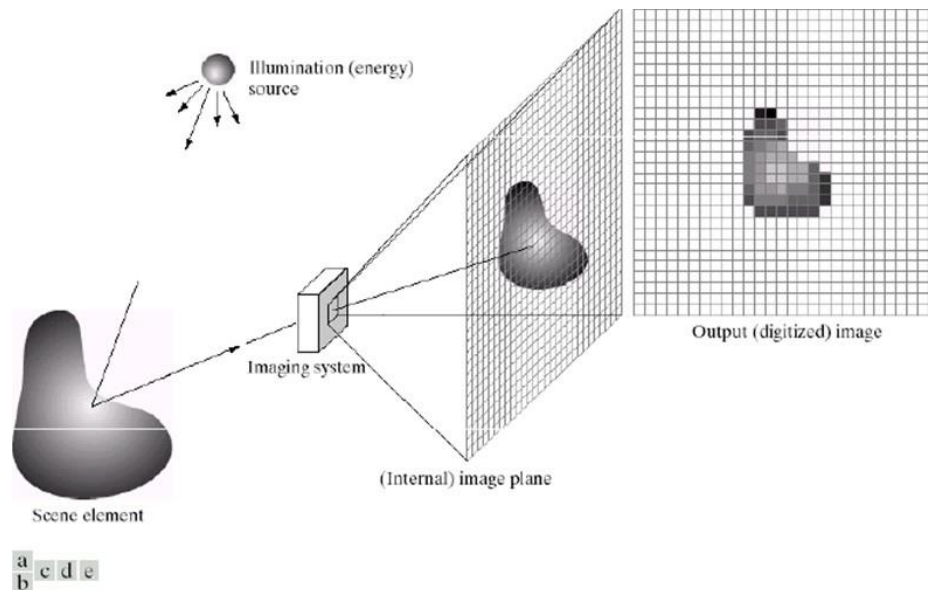
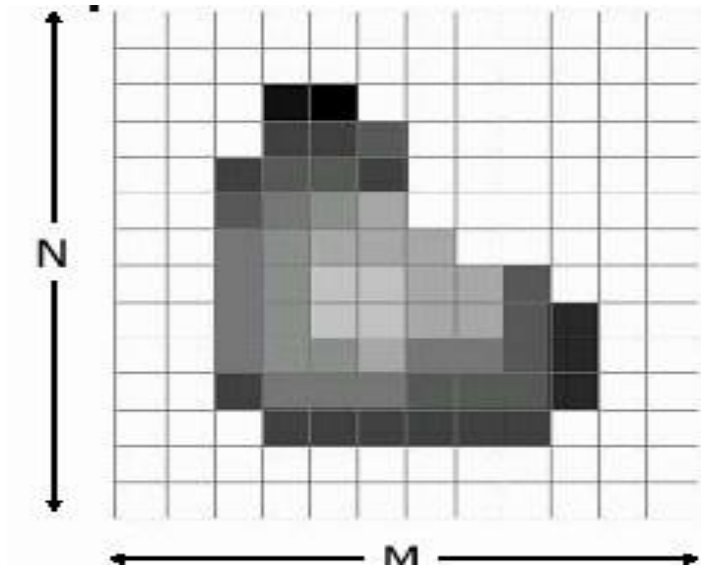


FIGURE 2.15 An example of the digital image acquisition process. (a) Energy ("illumination") source. (b) An element of a scene. (c) Imaging system. (d) Projection of the scene onto the image plane. (e) Digitized image.

Image size



- The number of gray levels typically is an integer power of 2

$$L = 2^k$$

- Number of bits required to store a digitized image

$$b = M \times N \times k$$

Find the number of bits to store a 128X128 image with 64 gray levels.

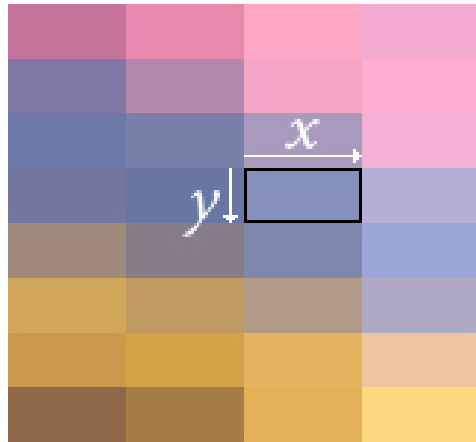
Given: $M = N = 128$

$$L = 64 = 2^k \Rightarrow k = 6$$

No. of bits = $M \times N \times k$

Number of bits to store image = $128 \times 128 \times 6 = 98304$ bits

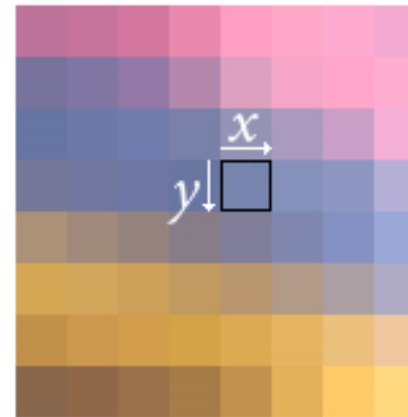
Aspect Ratio



Pixel Aspect Ratio

$$\text{PAR} = x/y = 2:1$$

x = pixel width
 y = pixel height



Pixel Aspect Ratio

$$\text{PAR} = x/y = 1:1$$

x = pixel width
 y = pixel height

Example : Find out the aspect ratio of the raster system using 8 x 10 inches screen and 100 pixel/inch.

Answer

We know that, Aspect ratio = Width/Height = $8 \times 100 / 10 \times 100 = 4 / 5$

Aspect ratio = 4 : 5

If an image has a height of 2 inches and an aspect ratio of 1.5, what is its width?

SOLUTION

$$\text{width} = 1.5 \times \text{height} = 1.5 \times 2 = 3 \text{ inches.}$$

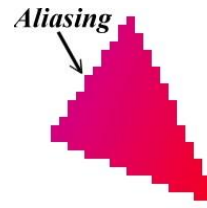
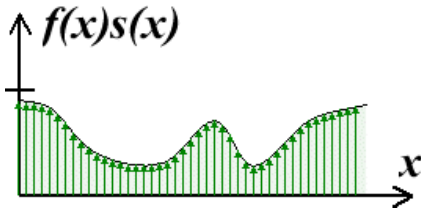
Aliasing

What Causes Aliasing?

- Aliasing happens if we do not sample the (continuous) image densely enough.
- We are not capturing the image details (high frequencies) with our discrete pixel grid.
- High frequencies arise at sharp edges or during perspective projections of texture images.

Dealing with Aliasing

- There are two ways of dealing with aliasing.
- 1) The first is to low pass filter our signal before we sample it.
 - 2) The second is to increase the sampling frequency



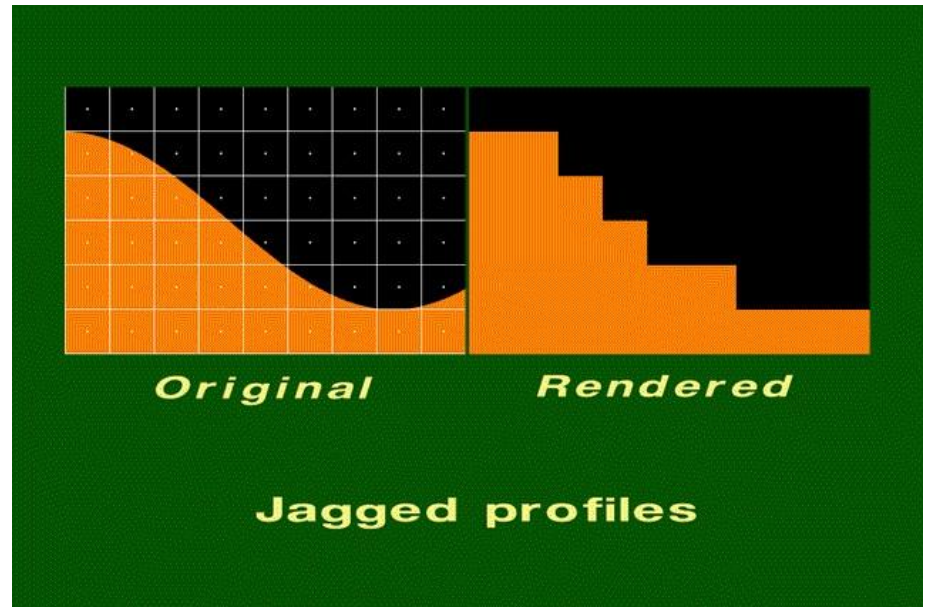
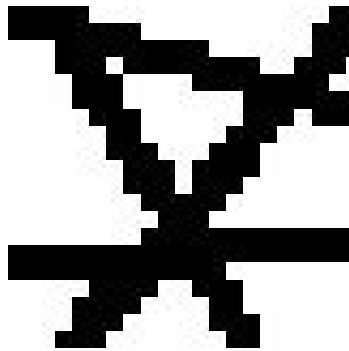
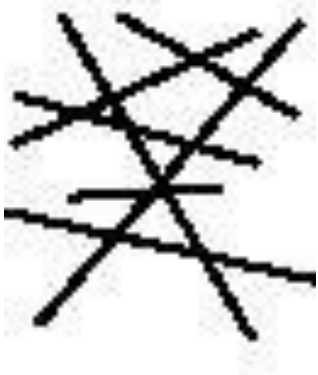
Major Types of Aliasing in computer graphics

- Spatial aliasing
 - Problems in individual images
- Temporal aliasing
 - Problems in image sequences (motion)

Spatial Aliasing

▪Jagged profiles:

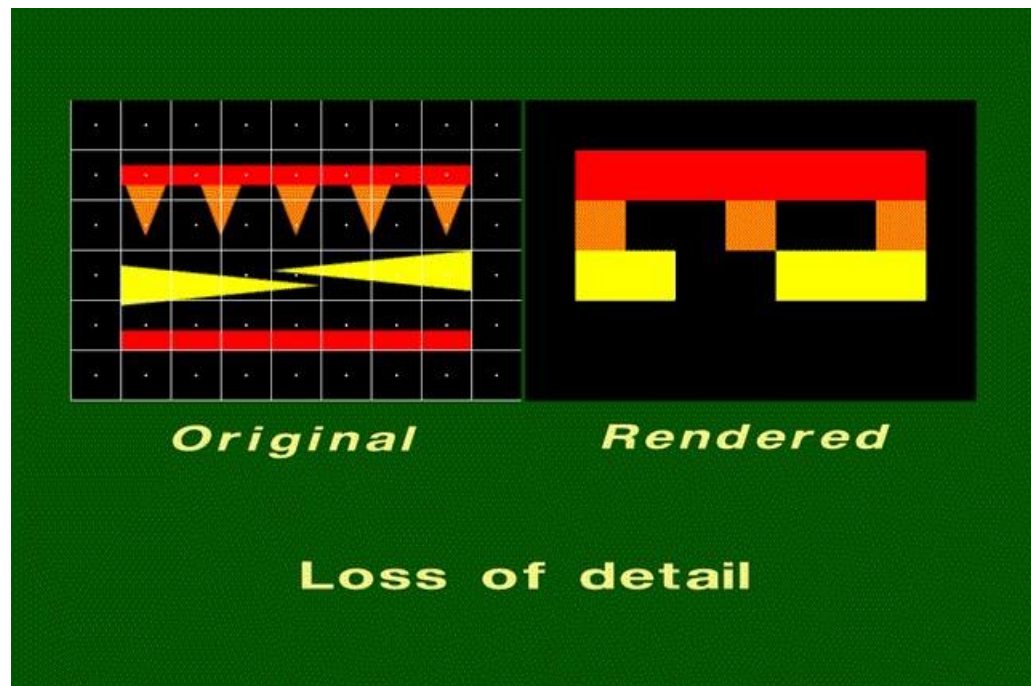
jagged (irregular – rough -zigzg) silhouettes are probably the most familiar effect caused by aliasing. Jaggies are especially noticeable where there is a high contrast between the interior and the exterior of the silhouette.



Spatial aliasing

- **Improperly rendered detail:**

In the rendered scene, one of the two red rectangles disappears entirely, and the other doubles in width. Two of the orange triangles disappear. Although the two yellow triangles are identical in size, one is larger than the other in the rendered image.



Spatial aliasing

▪ Disintegrating textures

The checkers should become smaller as the distance from the viewer increases. However, the checkers become larger or irregularly shaped when their distance from the viewer becomes too great.



Temporal aliasing

- **Strobing**

- Motion pictures are made from still images shown in rapid sequence. Provided there is sufficient continuity between the images and provided the sequence is shown fast enough, the central nervous system interprets the sequence as continuous motion. However, some technologies cannot process or carry data fast enough for sufficiently high frame rates. For example, viewing motion pictures by Internet connection generally necessitates a greatly reduced frame rate, making jerkiness clearly apparent.

- **Flickering**

- The **flicker** is **caused** by the fact that the lights strobe at the line frequency of the electrical power. If the frame rate of the **video** capture is close to that frequency (but not the same), then the effect becomes very noticeable. i.e Monitor refresh too slow, Frame update rate too slow
- Flicker occurs on CRTs when they are driven at a low refresh rate, allowing the brightness to drop for time intervals sufficiently long to be noticed by a human eye



Computer Graphics Applications

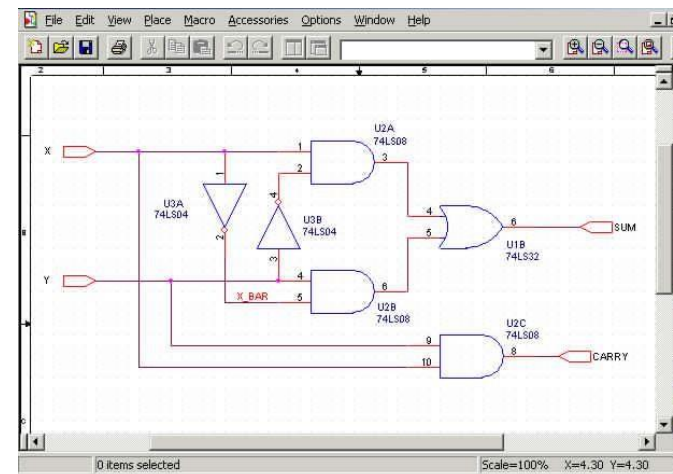
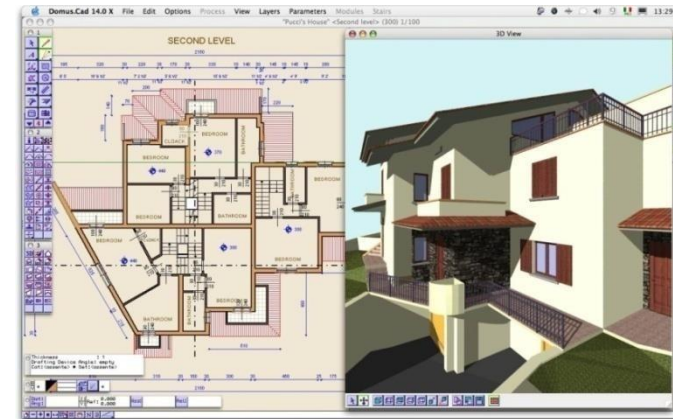
Computer graphics is used in every field. Some of the major application areas of its are:

1. Building Design and Construction

Computer graphics are very useful tool for generating architect's drawings and visualizing structures. Taking the data about the building, the computer can wake various image building from different angled. The architect can walk through" building, looking at image the computer.

2. Electronics Design

We know electronic devices like radio; television and computers are made up of many small electronic components. These parts are mounted on boards, called printed circuit design and interconnected by copper tracks on boards. Until few years ago the designed PCBS was done entirely by hand. Today this procedure is completely computerized. The engineer works on video screen and designs the PCB. Once the design is 'completed, the design system generates a hardcopy output.





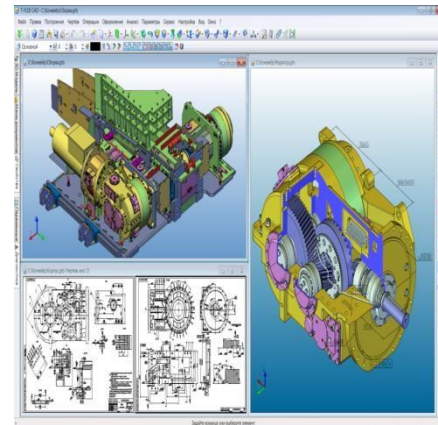
Computer Graphics Applications

3. Mechanical Design

We know machines are simply complex arrangement of mechanical parts. Today, constructive solid geometry technique (CSG) is used to design different parts on computer. Today before manufacturing any machine, engineers design that machines on computer first and with the help of that they design it.

4. Entertainment and Animation

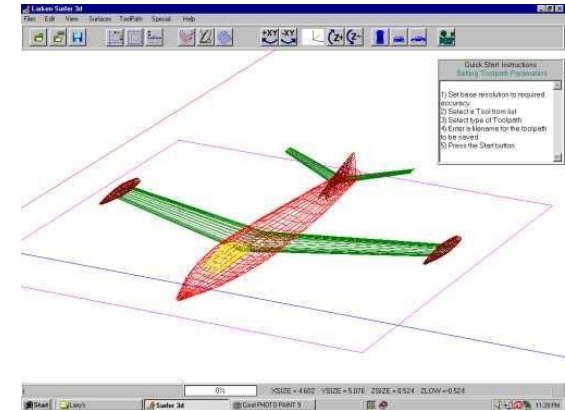
Computers are used in making cartoon animation films. It required tedious calculation for which computers are ideally suited.



Computer Graphics Applications

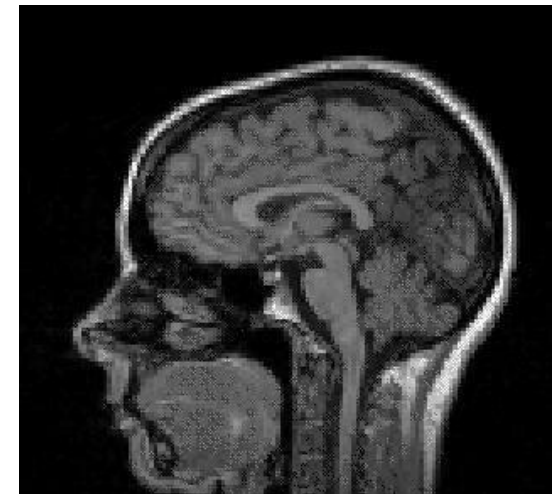
5. Aerospace Industry

Aircraft designers use computers to simulate the airflow over the wings. The air patterns around the wings are displayed on the graphics monitor. The patterns of the shock waves around the aircraft give the designer an idea about the area that may create unwanted turbulence.



6. Medical technology

Today, Computerized axial tomography (CAT) is used to compose the three-dimensional model of the brain by taking multiple X-rays of it. CAT scan is very useful in detecting various maladies like brain tumors. Computers are also useful to generate images of delicate organs. It is useful to study the effect of drugs on tissues. This treatment is available at Bombay Hospital, Bombay in India.



Computer Graphics Applications

7. Cartography

Computer graphics is used to produce accurate and schematic representation of geographical and other natural phenomena from measurement data. Examples include ' geographic maps, relief maps, exploration maps for drilling and mining, oceanographic charts, a weather maps, contour maps and population density maps. Eg google map/google earth



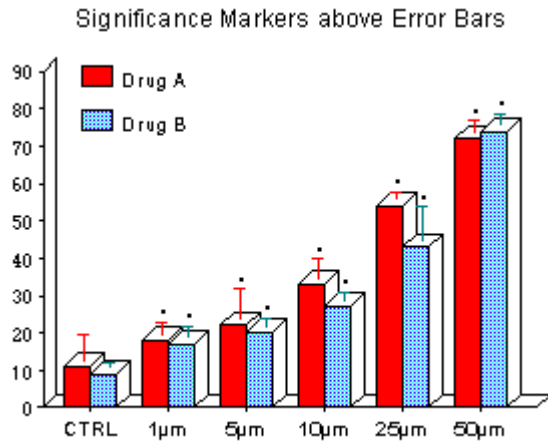
8. Art and Commerce.

Computer graphics is used to produce pictures that express a message and attract attention. Personal computers and teletext and videotext terminals in public places such as a museums, transportation terminals, supermarkets and hotels, as well as private homes, offer much simpler but still informative pictures that let users orient themselves, make choices, or even “teleshop” and conduct other business transactions. Finally, slide production for commercial, scientific, or educational presentations is another cost-effective use of graphics, given the steeply rising labour costs of the traditional means of creating such material.



Computer Graphics Applications

9-Presentation Graphics



visuals are processed

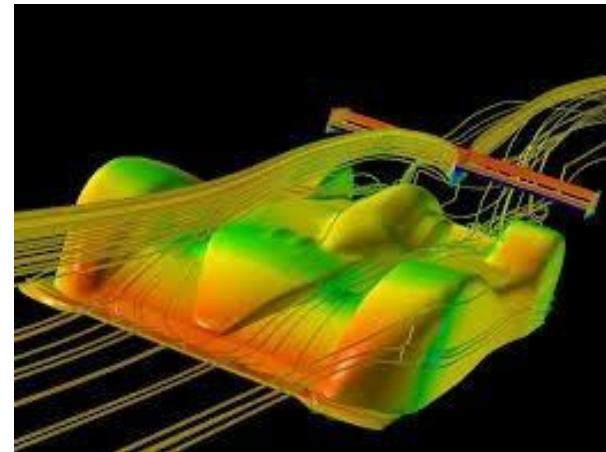
60,000 times faster

than text

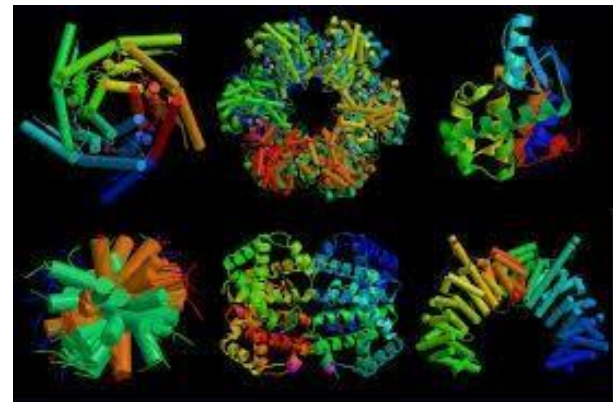
graphics

quickly affect our emotions
and our emotions greatly affect our decision making

10- Visualization



aerodynamics flow

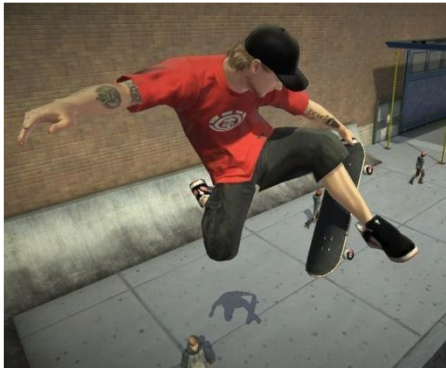


Protein Structure



Computer Graphics Applications

11-Games



12- Education and Training



Computer Graphics Applications

13- Graphical User Interface

